

Simulation Environment

Simulation Software

- Wokwi Online Simulator

Simulated Components

- Raspberry Pi Pico W
- HC-SR04 Ultrasonic Sensor
- Breadboard
- LED
- Buzzer

Programming Language

- MicroPython

Cloud Platform

- ThingSpeak

Simulation Procedure

1. Create a Raspberry Pi Pico W project in Wokwi.
2. Add the HC-SR04 sensor and connect it to the Pico W.
3. Write the MicroPython program.
4. Run the simulation.
5. Adjust the ultrasonic sensor distance to simulate different garbage levels.
6. Observe the serial monitor and ThingSpeak dashboard.
7. Capture screenshots for documentation.